

Vaishnavi Singh

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EDUCATION

Kalinga Institute of Industrial Technology

Bachelor of Technology in Computer Science (CGPA: 8.52/10)

Bhubaneswar, Odisha

Jul. 2023 – Jul. 2027

St. Joseph's Convent High School, Bankipur

Senior Secondary (Class XII) (80%) | Secondary (Class X) (92.4%)

Patna, Bihar

2020 – 2022

EXPERIENCE

Backend ML Intern

Oct. 2025 – Dec. 2025

Infosys SpringBoard (Remote)

- Engineered secure REST APIs and scalable ETL pipelines on Azure serving ML demand forecasting models; automated model selection across ARIMA, LightGBM, and LSTM – reducing manual evaluation effort by ~60%.
- Processed 50K+ multi-region time-series data points; built capacity optimization dashboards and presented risk indicators to leadership, directly influencing infrastructure planning decisions.

Microsoft Learn Student Ambassador (MLSA Lead)

Feb. 2024 – Mar. 2026

KIIT University

Bhubaneswar, Odisha

- Led 10+ technical workshops and hackathons with 2,000+ participants across AI/ML, Cloud, and Cybersecurity tracks; mentored 100+ student developers in adopting industry-grade tools and engineering practices.

PROJECTS

Enterprise Demand Forecasting & Capacity Optimization | Python, Flask, Azure, LightGBM, LSTM

- Developed a scalable ML forecasting system predicting CPU, storage, and user load across Azure regions; deployed REST APIs for real-time inference with an automated MLOps pipeline for CI/CD model updates.
- Reduced infrastructure over-provisioning by surfacing data-driven capacity recommendations; built stakeholder dashboards tracking forecast accuracy and risk indicators across distributed compute.

Distributed Rate Limiter as a Service | Python, Redis, FastAPI, token bucket algorithm, Docker

- Designed and implemented a distributed rate limiting service using the token bucket and sliding window algorithms; deployed via FastAPI with Redis-backed state sharing across multiple service instances.
- Benchmarked throughput under 10K+ concurrent requests; achieved consistent enforcement of per-client request quotas with sub-5ms overhead, supporting horizontal scaling via stateless API design.

Quad Bayer RAW Image Reconstruction Pipeline | Python, PyTorch, NumPy, OpenCV

- Built a deep learning ISP pipeline for Quad Bayer RAW demosaicing and noise reduction; benchmarked output quality via PSNR/SSIM metrics across noise levels targeting low-light mobile camera enhancement.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, System Design, Distributed Systems, Complexity Analysis

ML & AI: Machine Learning, Deep Learning, NLP, GenAI (LLMs, Prompt Engineering, RAG), PyTorch, TensorFlow, Scikit-learn, Hugging Face

Data & MLOps: Pandas, NumPy, ETL Pipelines, Flask, FastAPI, Docker, CI/CD, REST APIs, Azure, Power BI

Tools: Git, Jupyter, Google Colab, Streamlit, OpenCV, Figma

CERTIFICATIONS & ACHIEVEMENTS

Infosys Springboard (2025): Generative AI, Prompt Engineering, GPT-3 & OpenAI Models, Deep Learning, Computer Vision, NLP, Data Science, Agile Scrum

Cisco NetAcad: CCNA: Enterprise Networking, Security, and Automation, Data Science, Modern AI

HackVega 2025: Finalist, nationwide hackathon by CareerNet – top teams selected from pan-India submissions

AWS: Completed cloud computing labs covering core infrastructure, compute, and storage services

SOFT SKILLS

Leadership & Mentorship | Technical Communication | Cross-functional Collaboration | Agile Teamwork | Problem Solving | Adaptability